

Node operations and security framework

NODE-OPS-1.0 · review of github.com/AzielEliab · 2026-09-06 · Aziel Eliab

Bulletproof here means smallest public surface, named refuses, isolate-then-phoenix wait / re-seal, no auto-heal onto a poisoned ID, and public tunnels / sites that die with the pull. It does not mean an unbreakable host or a hostname that climbs back.

1. Fleet review (2026-09-06)

Public repos already point the right way: FragGate hashed registry and refuse ledger; corpus security headers; runtime thin MCP; qnm-node on 127.0.0.1 with APG, PHOENIX-LOCK wait / re-seal, no account resurrection, no auto-heal; ARK hosted-never-unlocks; azos/exec stub; MirageGrid hop stubs; AZMail smtp/login stubs. Gaps: BUILD-1.0 docs still say bearers default-off (amend per QNM-WP-1.0: process ON, public rollup); cell-of-25 dual-bridge rotation not fleet-wired; Q×act seal not yet the only legal success path on every fraggate_call.

2. Node classes

Class	Fleet	Public surface allowed
Face	azieleliab, corpus, godlock	Pages, cite, llms, software mirror while the site exists. Pull site / revoke token / drop Worker / kill DNS → the public face dies with the pull. Phoenix does not restore it.
Door	fraggate, aziel-runtime	Thin MCP + fraggate_call. GET /v1/mesh never enables.
Fabric	qnm-node	Loopback API only. Not on public faces.
Survival	azieltether	Counted download. Prefer-central. Not a VPN.
Overlay	azos, azhub, azinterface	Status / place / cycles. No remote shell.
Harness	azbot	Skill markdown. Queue publish. No browser login.
Vault	ark, EmbryoLock	Local only. Hosted unlock/scorch stub.
Assign	miragegrid	assign live. mesh/vpn-hop stub.
Side-net	aznet, azbrowser, azmail	Verify / cite-search / classify. Harvest/login/smtp stub.
Engine	Locks, CLCE, Z-Solver, receipts, GATE	Listed live ops only.

3. Attack surface law

- One inbound door. FragGate or loopback. /p/{slug}/{op} is proxy, not exec.
- Allowlist ops only. Unknown slug = FG-HALLUC-TOOL. Stubs stay refuse.
- No credentials in process state. token_present only. No browser login.
- No public enable switch. GET does not change mesh state.
- qnm-node binds 127.0.0.1. Faces do not proxy that port.
- Public tunnels and sites DIE WITH THE PULL. A Cloudflare Tunnel lives on token + DNS name + account. Pull site / revoke token / drop Worker / kill DNS → cloudflared has nowhere legal to land. Restarting cloudflared is operator kit, not this contract; it fails if credential or hostname is gone.
- No vault on the Worker. No auto-heal to a spent mesh_id. No public hostname restore.
- No account resurrection. Memorial keeps the spend.
- Identity lock on export. Publish only via official API + AZBOT_PUBLISH=1.
- Local HTTP keeps nosniff, DENY frames, no-referrer, camera/mic/geo off, tight CSP.

4. Phoenix loop (wait / re-seal — not public restore)

live → APG miss or tamper → isolate (cut tethers, spend mesh_id) → memorial → PHOENIX-LOCK wait / re-seal (local, no controller hunt). Stop there.

- Phoenix is wait / re-seal after poison or isolation. It is not “bring the .uk / public node back.” It does not restore a public hostname, Cloudflare tunnel, Worker, or public rollup.
- This is not auto-heal. The old ID stays spent. No controller hunt.
- Sites pulled → public rollup down. Local node may keep verifying and appending. Mesh does not climb back onto the public hostname by itself.

- A later local assign of a new mesh_id is a separate operator act on the local cell — not Phoenix reseating the public name, not a failsafe that heals the pulled site.
- Phoenix does not un-scorch a vault.
- Poison does not forward across cell edges.
- Three missed heartbeats = suspect. Suspect + APG hit = isolate.
- Bridge poison rotates both bridges before inter-cell traffic.
- Held publish queue does not flush across phoenix without a new operator act.
- Phoenix does not un-pull a site, re-issue a tunnel token, or rewrite DNS.

5. Per-class refuse (minimum)

Class	Must refuse
Face	Node Gate, IP panel, vault unlock, mesh enable, hostname resurrection
Door	Invented slugs, stub ops, runtime_run as agent default
Fabric	cell_full join, leaf as inter-cell path, spent ID reuse, public bind
Survival	vpn / arm / mesh-join stubs; opening a public mesh because central is down
Overlay	azos/exec; exec of catalog engines from Interface
Harness	live post without PUBLISH=1; browser login
Vault	hosted unlock / encrypt / scorch / wipe
Assign	mesh, vpn-hop, hop, tunnel
Side-net	harvest, login, smtp send, invented visits, deanonymize
Engine	any op not on the 1.6.13 allowlist

6. Receipts

Every node writes AZL-LEDGER kinds: boot, heartbeat, refuse, isolate, phoenix, assign, rotate, tether_cut, hold. No model sentence. No token body.

7. Close tests

- Public faces have no enable switch and no vault unlock.
- Door stubs refuse and the refuse is a receipt.
- qnm-node is not reachable off loopback in default config.
- Spent mesh_id rejected on later local cell join. Phoenix wait has no outbound hunt packet.
- Phoenix does not restore a public hostname. Pulled sites stay down (die with the pull).
- Held queue does not flush across phoenix without a new act.
- ARK hosted unlock remains stub. Cell does not halt when one leaf isolates.

Does not add a Node Gate to “see security.” Does not turn MirageGrid into a VPN. Does not publish exploit recipes. Public identity: Aziel Eliab only.

If the files hold, the name was never the point.